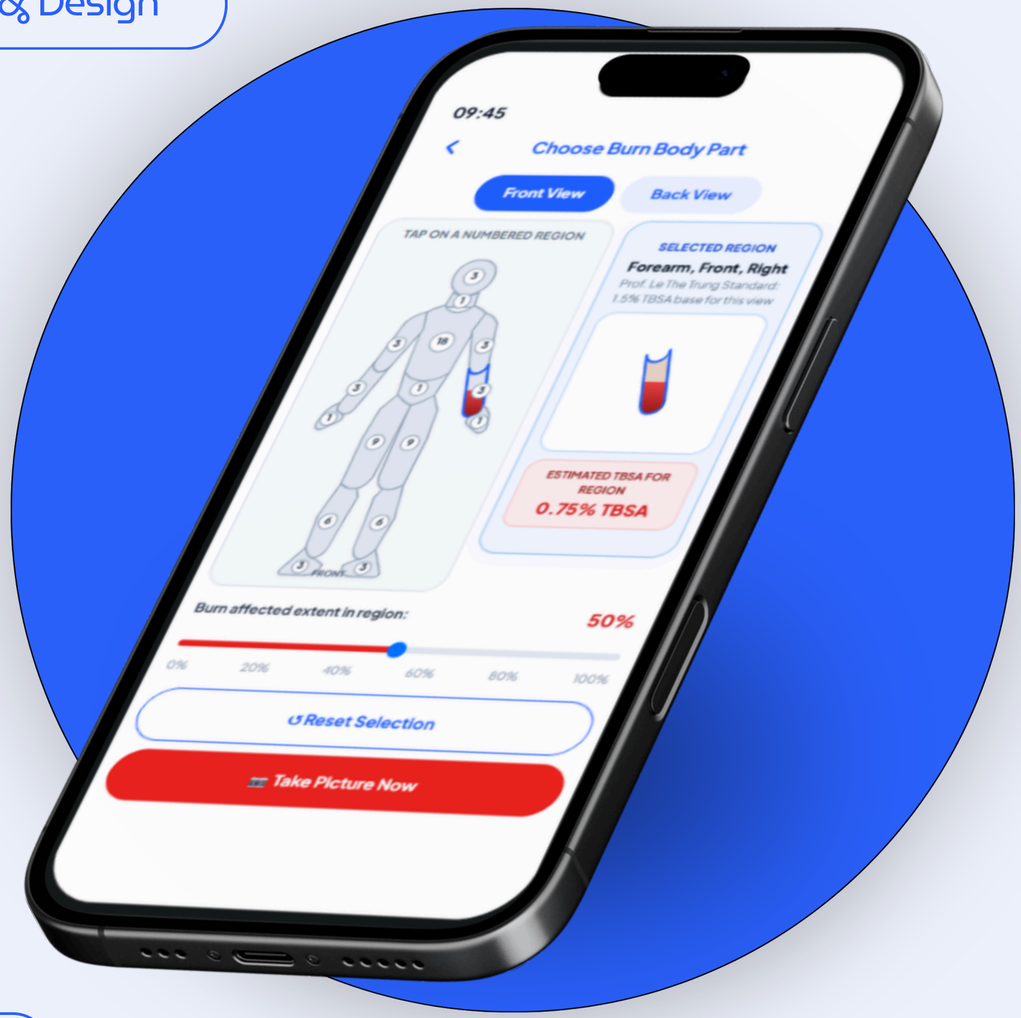
UEH UNIVERSITY

INSTITUTE OF INTELLIGENT & INTERACTIVE TECHNOLOGIES

Information System Analysis & Design

BURN AID

Burn assessment & First-aid guidance Information system



GROUP 3

Đức Huy - Phương Thảo - Quốc Túy - Khánh Uyên - Lưu Vũ

Burn Aid

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1 Overview

2 System modeling

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Background & Problem

- Burn injuries are a **public health concern** that can occur at home, in the workplace, and in other everyday settings.
- Before medical help arrives, patients must j**udge severity and give first aid**.
- AI and computer vision can **classify burn depth and outline wound**s from photos.

How might we **provide preliminary burn assessment** to whom experience burn before medical help is available **to support decision-making**.

Proposed Solution

Burn Assessment and First Aid Guidance Information System

1. Upload burn photo + short survey
2. AI classifies depth, segments burn
3. First-aid guidance

Objective

Develop and evaluate an integrated prototype that supports early burn assessment and access to reliable first-aid information

Scope

In scope

- Adults aged 18+
- Image + survey assessment
- First-aid guidance and warnings

Out of scope

- Diagnosis or treatment
- Children under 18
- Ambulance or location sharing

Stakeholders

Stakeholders	Needs	Influence Level
Department of Health, public health organizations	• Reduce the burden on hospital emergency departments from minor burn cases • Promote standardized public first-aid practices.	High
Patient and Family members	• Receive appropriate assessment and first-aid guidance for minor burns and timely emergency assistance for severe burns.	High
Medical advisor	• Ensure that first-aid guidance follows medical standards and validate the reliability of the AI model.	High
IT Staff	• Ensure stable system performance and protect users' personal image data.	Medium
Emergency services	• Receive emergency phone calls initiated by users via the device's native dialer when recommended by the app.	Low

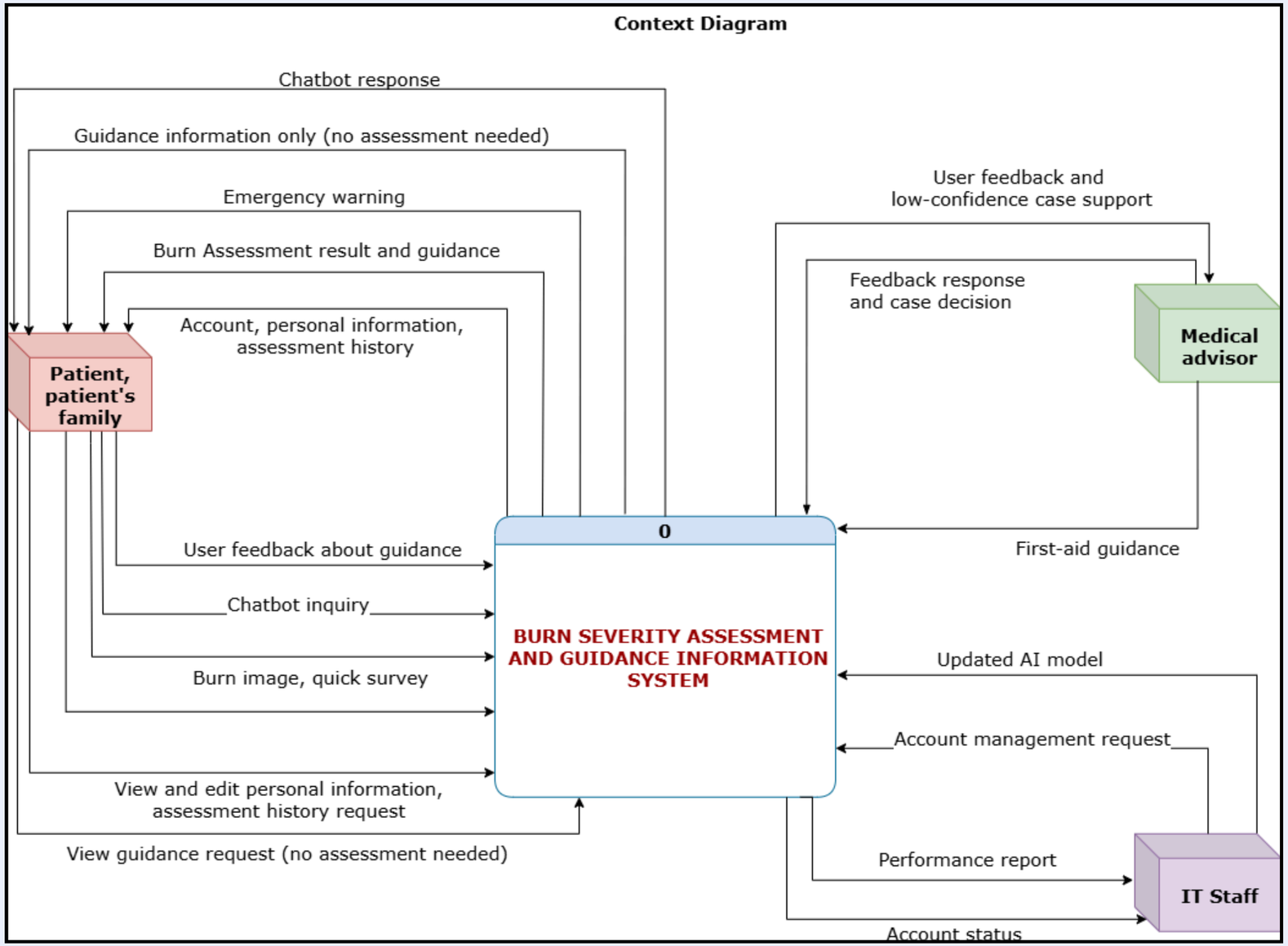
User roles and requirements

User roles	Description
Patient and Family member	• Primary user group of the system. • Use it when they need a quick preliminary assessment of a burn, suspect a burn injury, or require appropriate first-aid guidance.
Medical Advisor	• Reviewing and managing professional first-aid content within the system.
IT Staff	• Managing the AI-based assessment models and maintaining the system. • Ensure that the system functions, user accounts, and model-generated assessments operate appropriately and effectively.

Context Diagram

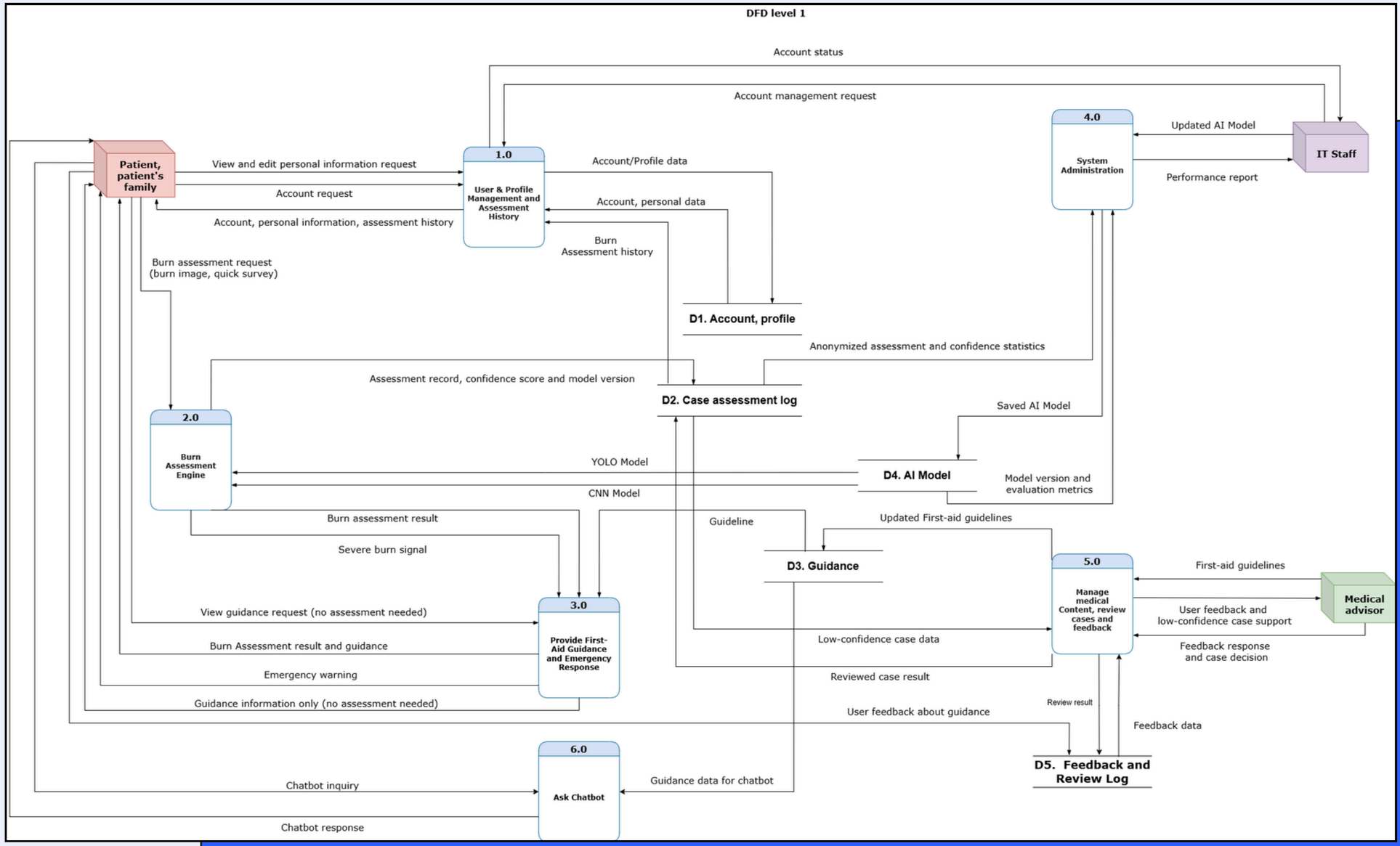
3 external entities:

- Patient and Family member;
- Medical Advisor;
- IT Staff



Burn Aid

DFD *level 1*



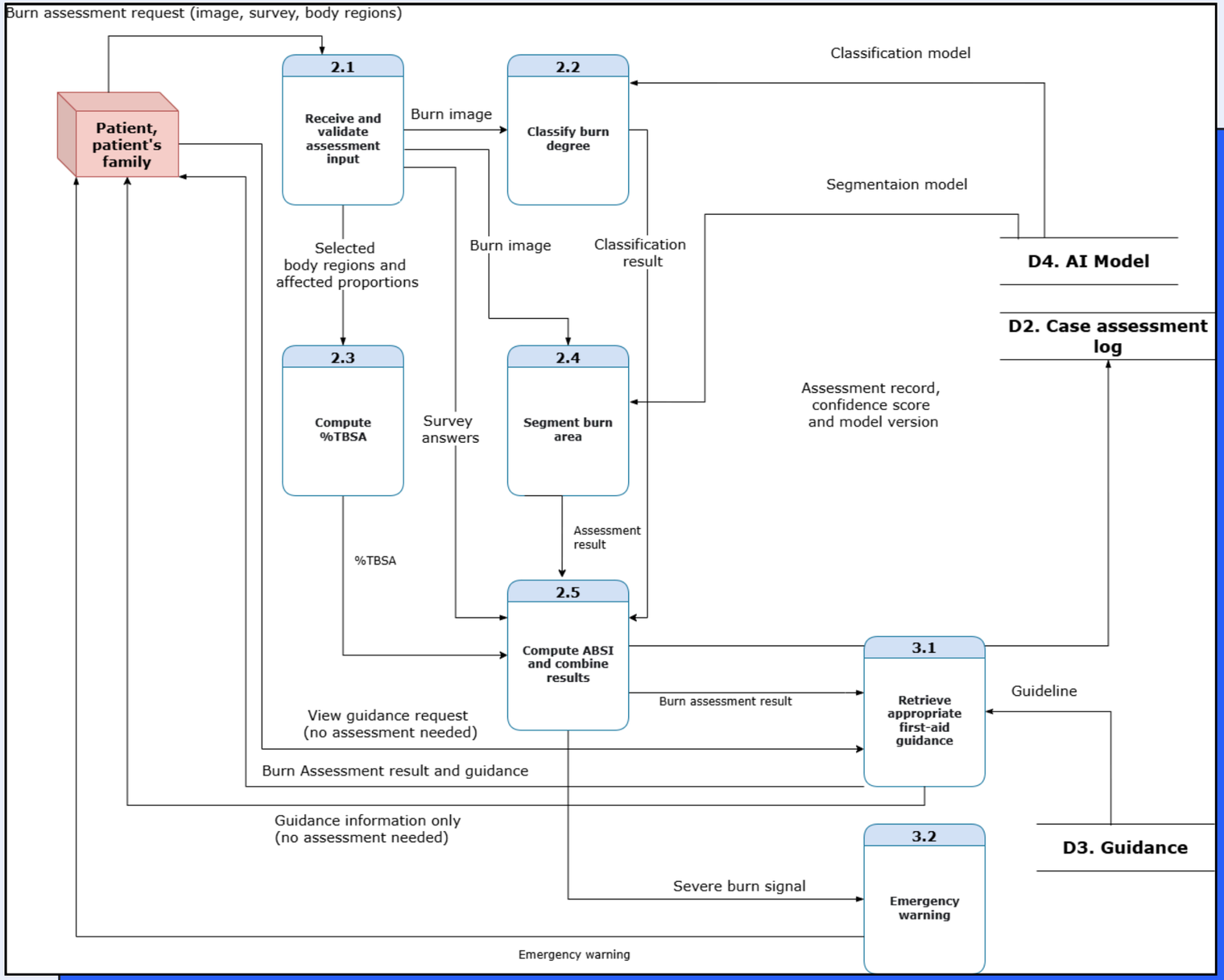
Burn Aid

DFD level 2

Process 2.0 has
5 sub-processes.

Processes 3.0:

- Guidance, emergency warning,
- Case review.

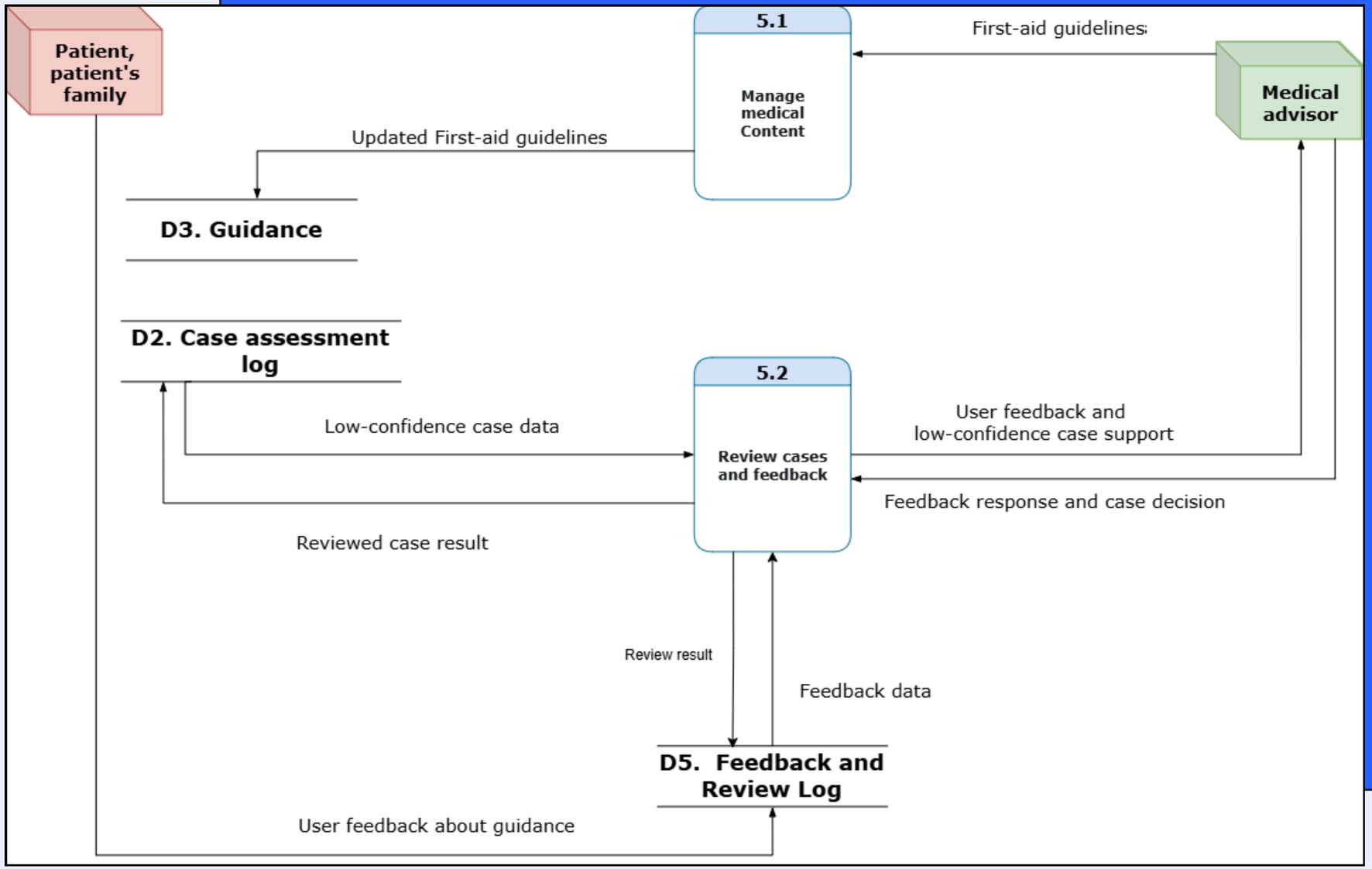


Burn Aid

DFD level 2

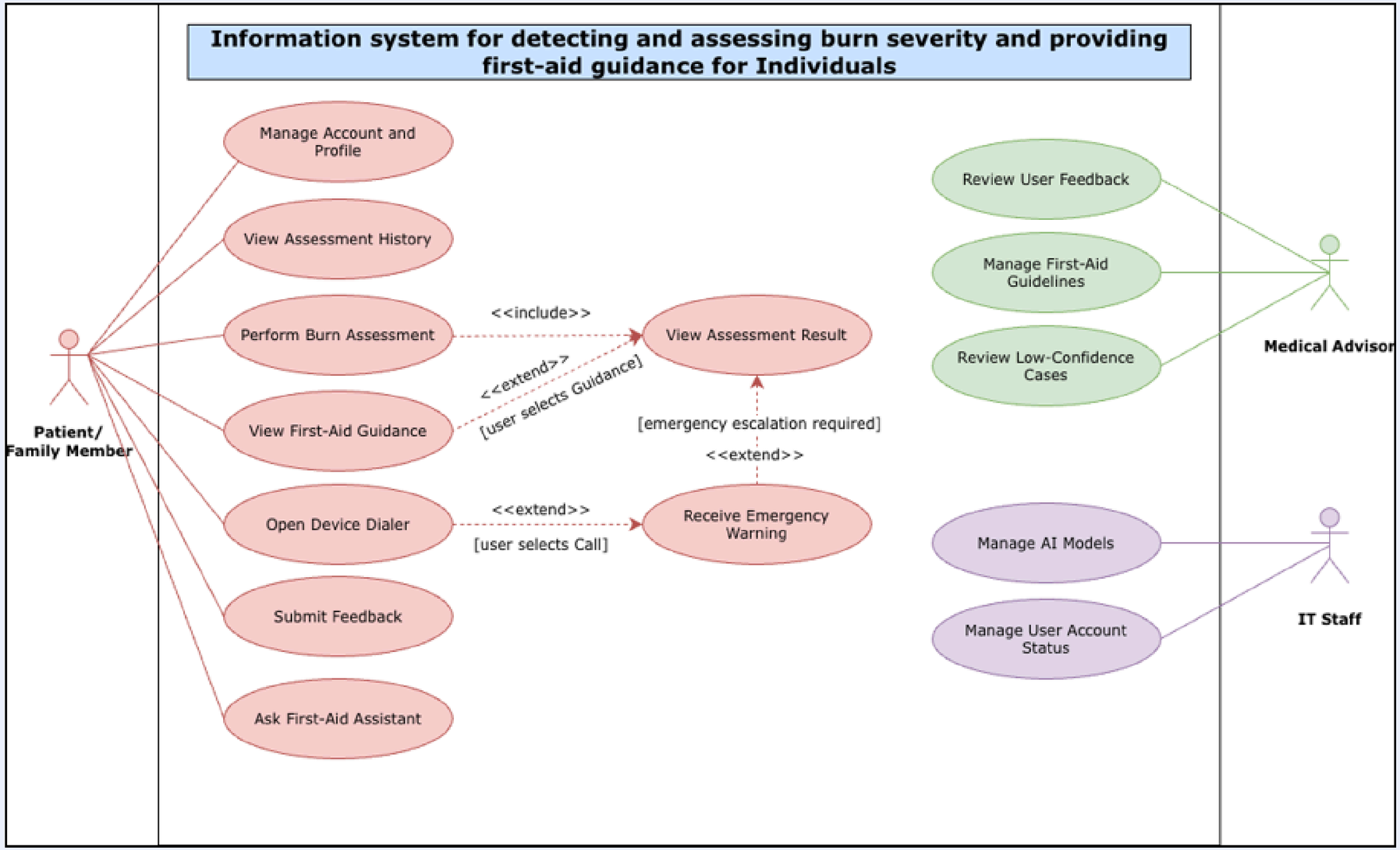
Processes 5.0:

- Manage medical review,
- Review cases and feedback.



Burn Aid

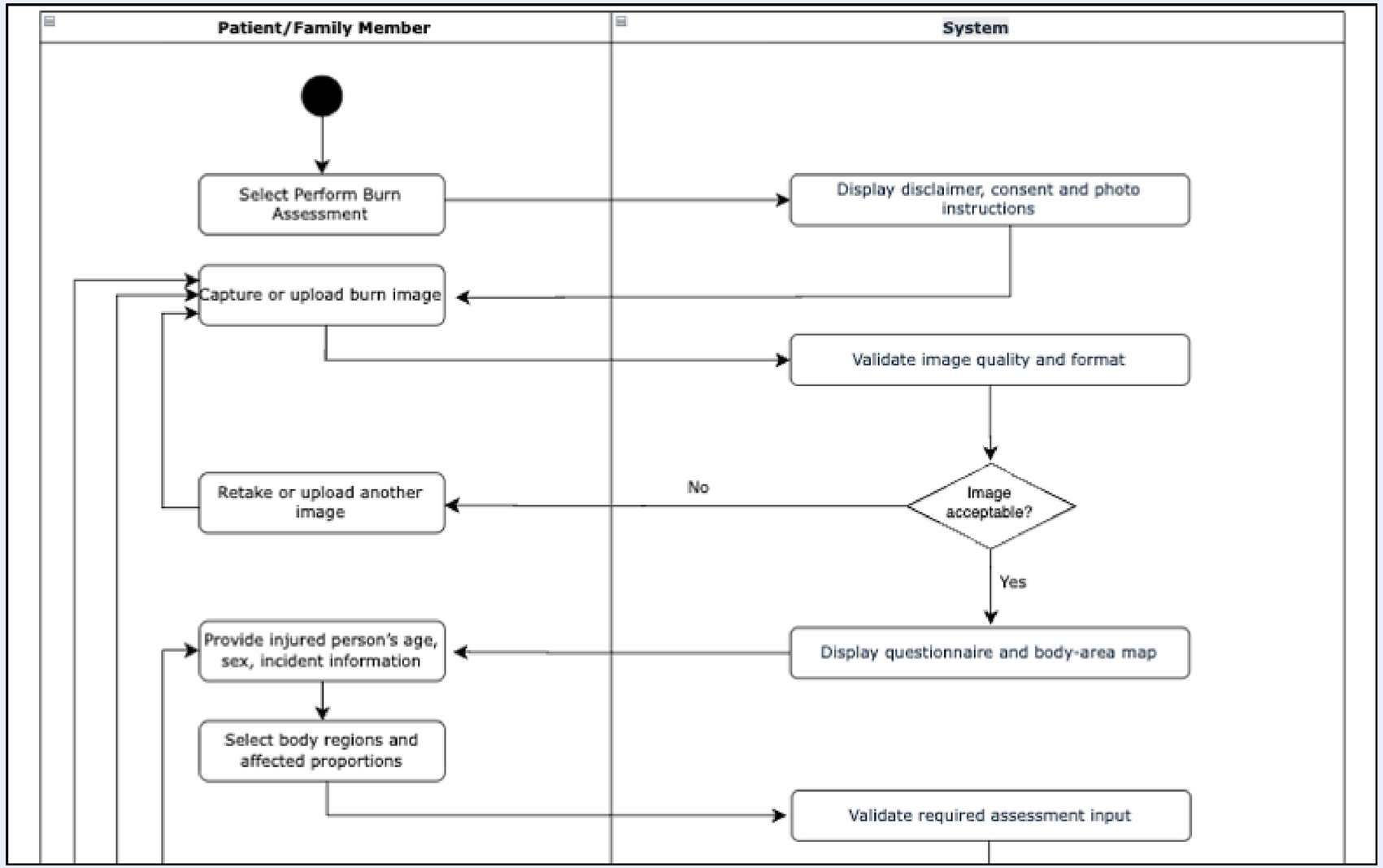
Use Case Diagram



ID	Use Case
UC-01	Manage Account and Profile
UC-02	View Assessment History
UC-03	**Perform Burn Assessment**
UC-04	View Assessment Result
UC-05	View First-Aid Guidance
UC-06	Receive Emergency Warning
UC-07	Open Device Dialer
UC-08	Submit Feedback
UC-09	Review User Feedback
UC-10	Manage First-Aid Guidelines
UC-11	Review Low-Confidence Cases
UC-12	Manage AI Models
UC-13	Manage User Account Status
UC-14	Ask First-Aid Assistant

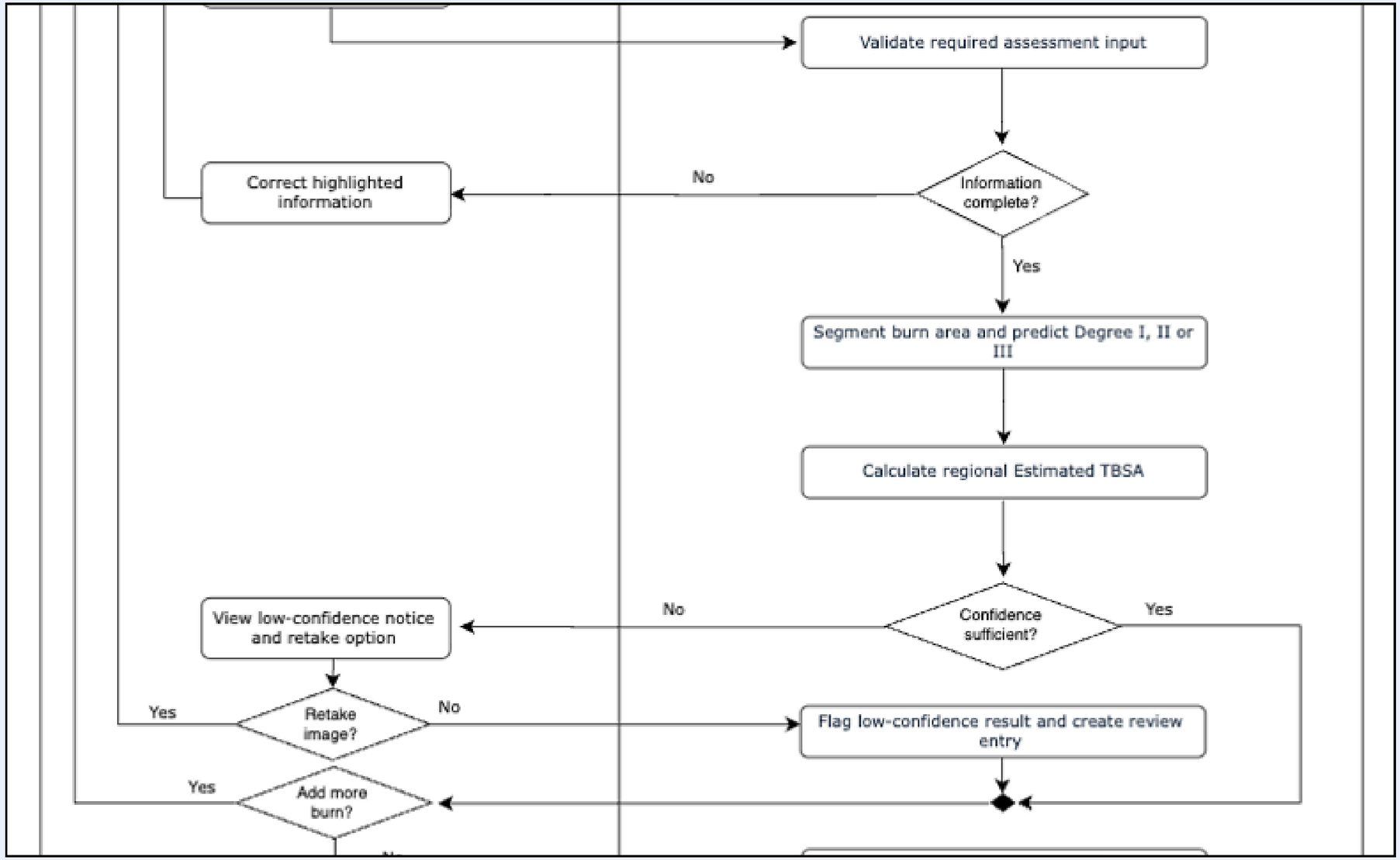
Activity Diagram for UC-03

Perform Burn Assessment



Activity Diagram for UC-03

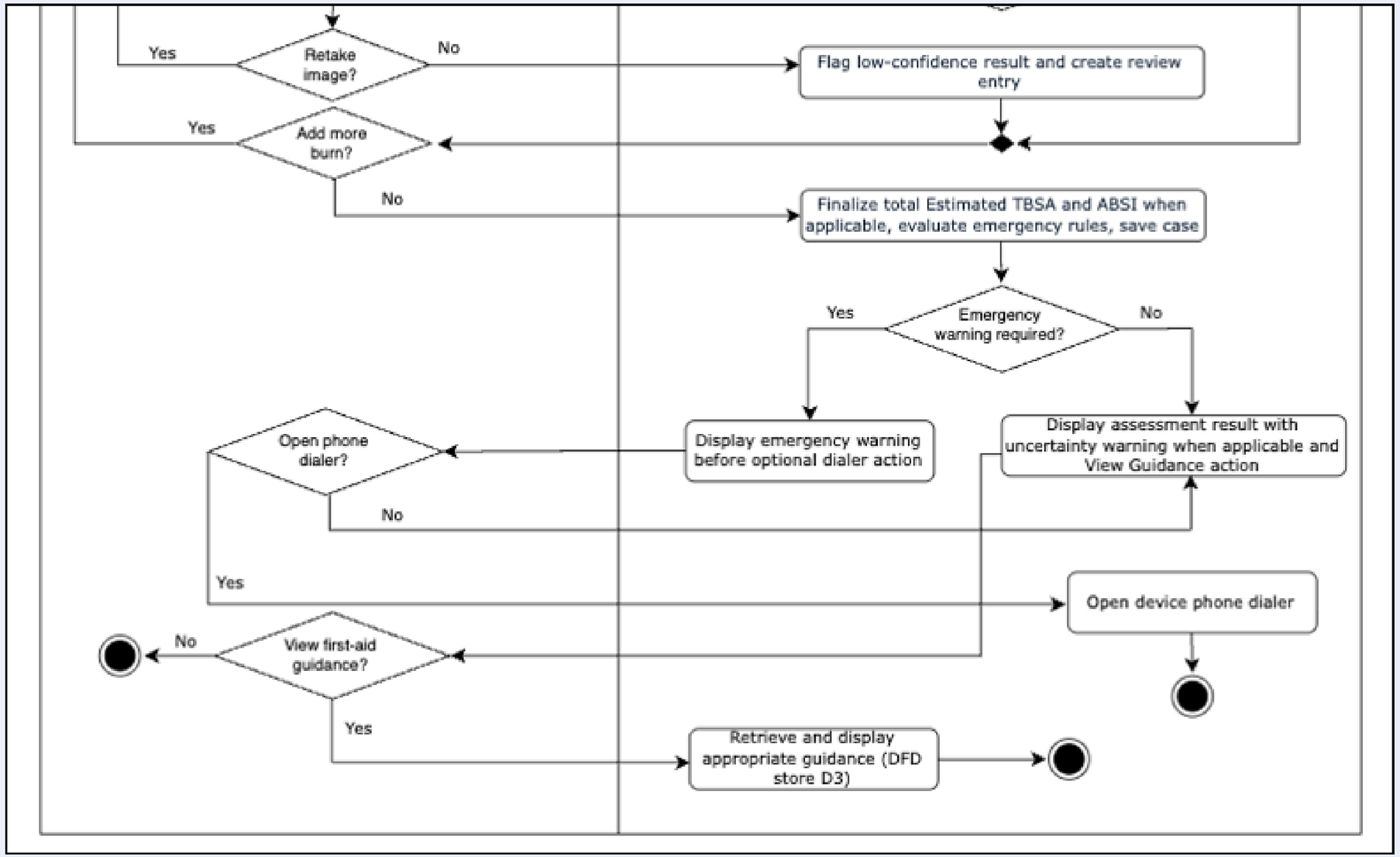
Perform Burn Assessment



Burn Aid

Activity Diagram for UC-03

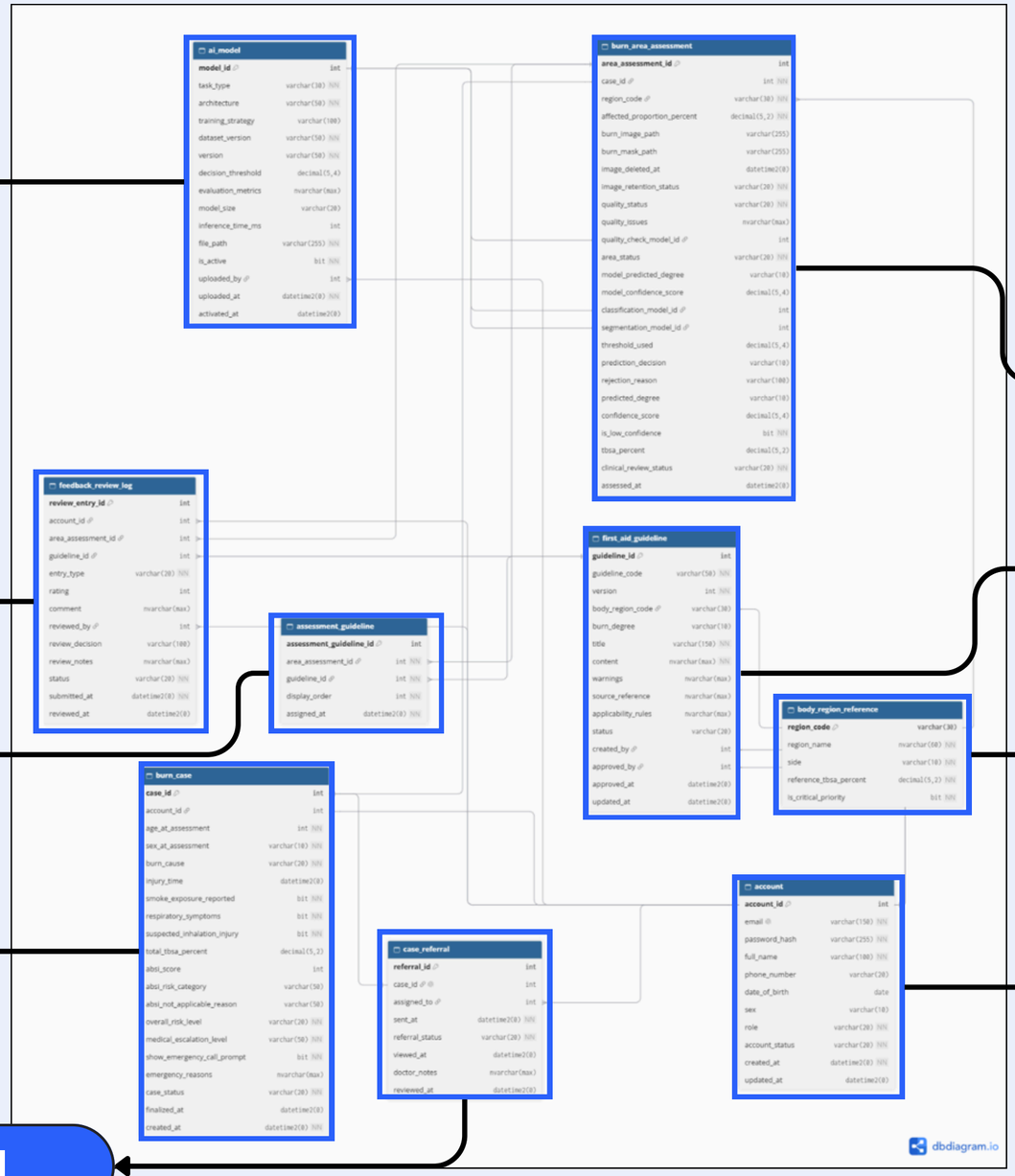
Perform Burn Assessment



Relational Schema

Table	Relational Schema
Account	Account(**account_id**, email, full_name, role, account_status, ...)
Burn_case	Burn_case(**case_id**, account_id (FK), age_at_assessment, burn_cause, case_status, ...)
Burn_area_assessment	Burn_area_assessment(**area_assessment_i**d, case_id (FK), region_code (FK), predicted_degree, confidence_score, tbsa_percent, ...)
Body_region_reference	Body_region_reference(**region_code**, region_name, side, standard_tbsa_percent, ...)
First_aid_guideline	First_aid_guideline(**guideline_id**, created_by (FK), guideline_code, version, title, status, ...)
Assessment_guideline	Assessment_guideline(**assessment_id**, area_assessment_id (FK), guideline_id (FK), display_order, ...)
AI_model	AI_model(**model_id**, task_type, architecture, version, is_active, ...)
Feedback_review_log	Feedback_review_log(**log_id**, case_id (FK), area_assessment_id (FK), guideline_id (FK), account_id (FK), entry_type, status, ...)
Case_referral	Case_referral(**referral_id**, case_id (FK), assigned_to (FK), referral_status, viewed_at, ...)

ERD



AI_model

Feedback_review_log

Assessment_guideline

Burn_case

Case_referral

Burn_area_assessment

First_aid_guideline

Body_region_reference

Account

Burn Aid

Model Development

Objective

Support pre-hospital first aid for thermal burns by classifying burn depth and locating suspected burn regions.

Burn depth clasification	**ConvNeXtV2 Nano · 224 × 224 RGB**	Image-level probabilities for four classes: First-degree, Second-degree, Third-degree, and None (no burn)
Burn region segementation	**U-Net++ with ResNet-34 · 320 × 320 RGB**	Binary burn mask and visual overlay identifying suspected burn regions
Clinical assessment	**Independent Clinical Rule Engine**	Estimated %TBSA, risk level, and emergency alerts

Inference flow: *Image → Classification → If burn detected: Segmentation → Clinical assessment*

Classification model

Dataset

4,686 RGB images across four relatively balanced classes. ***Observed split:*** 83.40% train / 9.86% validation / 6.74% test

Class	Train	Validation	Test	Total
First-degree	1,076	110	73	1,259
Second-degree	1,021	97	83	1,201
Third-degree	834	140	81	1,055
No Burn (None)	977	115	79	1,171
Total	3,908	462	316	4,686

Classification model

Data cleaning and preprocessing

Cleaning & labels

- Crop and visually filter images in Roboflow
- Manually standardize the four class labels
- Screen duplicates using pHash similarity
- Check image files and single-label validity

All splits: RGB → tensor → ImageNet normalization

Standardize image quality and labels before fine-tuning

Image transforms

- Train: RandomResizedCrop to 224 × 224
- Horizontal / vertical flips; rotation ±15°
- ColorJitter for lighting variation
- Validation / test: fixed resize to 224 × 224

Classification model

Training setup

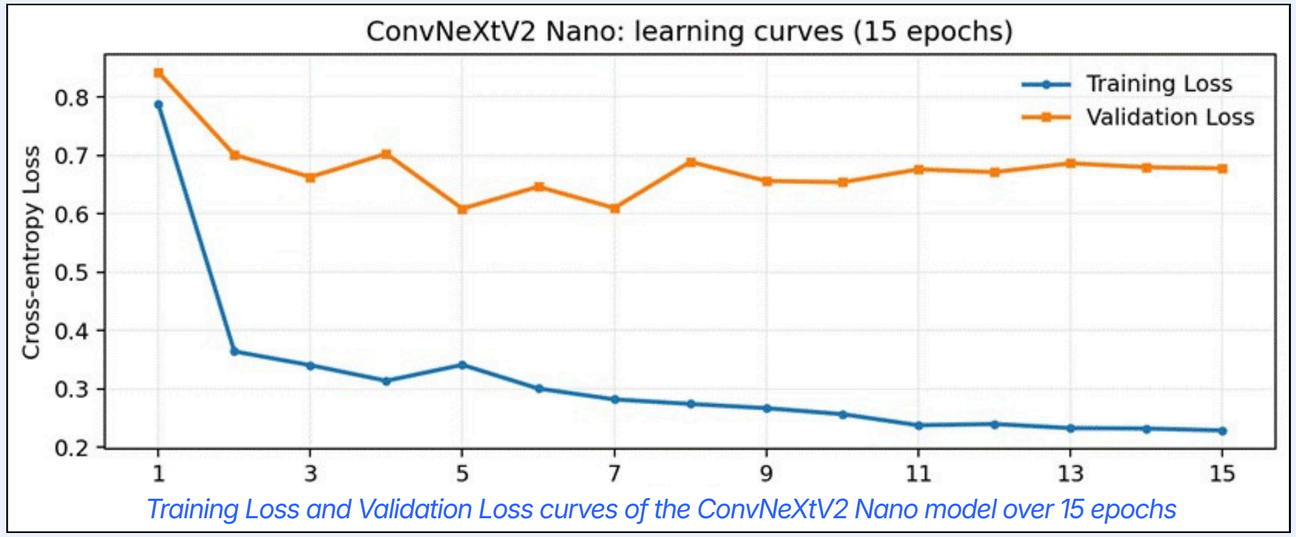
the ConvNeXtV2 Nano architecture via the timm library, initialized with pretrained weights convnextv2_nano.fcmae_ft_in22k_in1k

Setting	Configuration
Architecture / initialization	ConvNeXtV2 Nano · 14.99M parameters · pretrained
Training budget	15 epochs · Batch size 64
Optimizer	AdamW · Weight decay 0.05
Learning rates	Backbone: 2×10^{-4} | Head: 2×10^{-3}
Scheduler	OneCycleLR · pct_start = 0.15
Loss / regularization	Cross-entropy · Label smoothing 0.05 · Dropout 0.4

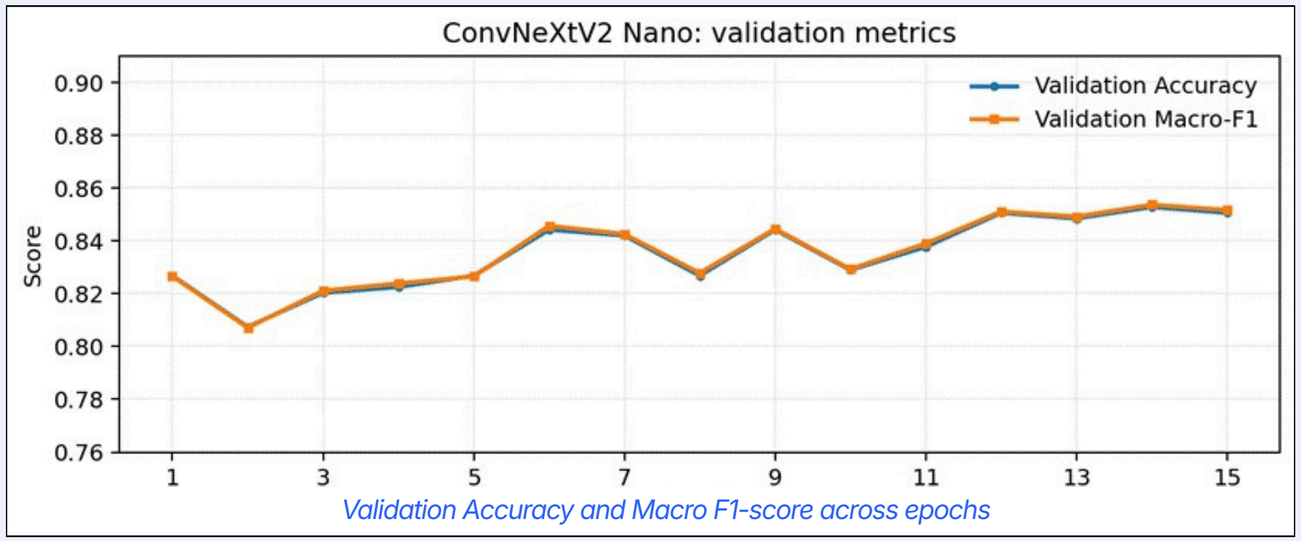
Checkpoint selection: highest validation Macro-F1

Training Progress

Lower training loss does not guarantee better validation performance



Training Loss and Validation Loss curves of the ConvNeXtV2 Nano model over 15 epochs



Validation Accuracy and Macro F1-score across epochs

Classification model

EPOCH 14

Selected checkpoint

85.37% Macro-F1

85.28% Accuracy

Training loss

0.7878 → 0.2285

Validation loss fluctuates; monitor overfitting.

Classification model

Results on test split

Best checkpoint evaluated on 316 held-out images

85.76%
Test accuracy

85.59%
Test Macro-F1

Class	Precision	Recall	F1-score	n
First-degree	85.00%	69.86%	76.69%	73
Second-degree	88.31%	81.93%	85.00%	83
Third-degree	73.00%	90.12%	80.66%	81
No Burn (None)	100.00%	100.00%	100.00%	79
Macro average	86.58%	85.48%	85.59%	316

First-degree burns have the lowest recall; Third-degree precision is 73.00%.

Classification model

Interpretation and limitations

Internal test performance does not establish real-world reliability

What the results show

- Macro-F1 reaches 85.59% on the internal test.
- First-degree recall is lower than other classes.
- No Burn is perfect on this split only.

What remains uncertain

- Healthy skin was occasionally classified as First-degree outside the dataset.
- Image provenance and license records are incomplete. Duplicate screening does not guarantee a leakage-free split.

Evaluate independent image sources, inspect First-degree errors, and strengthen traceability and duplicate checks across splits.

Segmentation model

Dataset

Combine manually annotated masks, quality-checked pseudo-masks and healthy skin

Mask source	Train manifest	Train used	Validation	Test
Manual polygon	300	300	21	13
QC pseudo-mask	921	921	0	0
None / background	977	879	0	0
Total	2,198	2,100	21	13

98 None images held out to monitor false-positive masks

300 manual training records ≈ 100 original image groups with augmentation variants. Validation and test contain only manual masks; both sets are small.

Segmentation model

Mask preparation and model output

Standardize image quality and labels before fine-tuning

Build training targets

- Manual polygons → binary masks
- Baseline U-Net++ → pseudo-masks
- Auto-QC: confidence, area, consistency
- Reject or isolate unreliable pseudo-masks

Preprocessing & predict

- RGB images resized to 320 × 320
- ImageNet normalization
- Nearest-neighbor mask resizing
- SHA-256, pHash and group checks

- The mask identifies suspected burn regions; it does not assign burn depth per pixel.
- Empty or skipped annotations must not be treated as healthy skin labels.

Output: one-channel prediction → binary mask → visual overlay

Training setup

Segmentation model

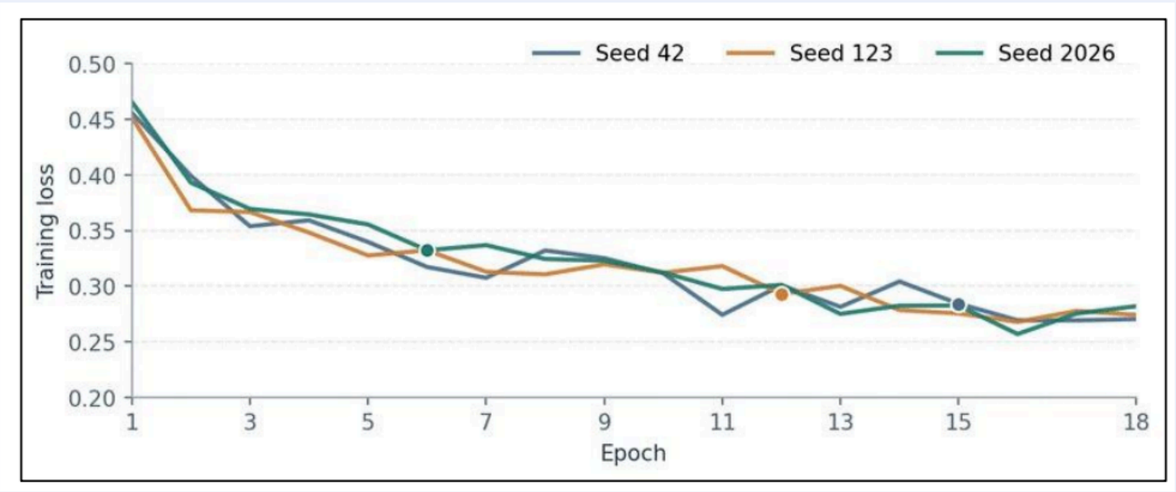
Technical Parameter	Configured Value
Model Architecture	U-Net++ (Nested U-Net), Encoder ResNet-34
Initial Weights	ImageNet Pretrained Weights
Input, Output Dimensions	320 × 320 RGB, 1-channel binary logits
Batch Size & Steps	Batch size = 8; 80 steps/epoch (640 samples/epoch)
Maximum Epochs	18 epochs
Loss Function	0.5 × BCEWithLogits + 0.5 × SoftDice
Source Sampling in Train	25% manual, 55% pseudo, 20% None
Checkpoint Criterion	SelectionScore = ValidDice - 0.15 × NegativeFPR
Best Checkpoint	best_random_curriculum_seed2026.pt (Epoch 6)

640 sampled images per epoch, drawn with replacement

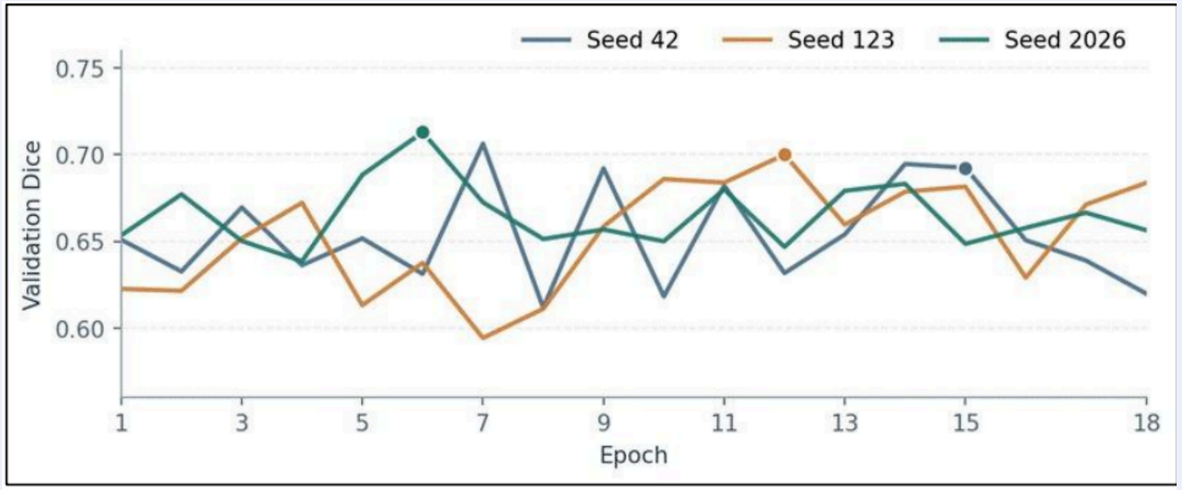
Training Progress

Random Curriculum trained 18 epochs across seeds 42, 123 and 2026

Segmentation model



Training Loss curves by epoch for Random Curriculum across three seeds (42, 123, 2026)



Validation Dice variation by epoch across three seeds, with dots marking selected checkpoints

SEED 2026

Training loss

0.4656 → 0.2819

Validation Dice fluctuates

65.34% → 65.64%

Lowest training loss

≠

best checkpoint

Segmentation model

Checkpoint selection and false positives

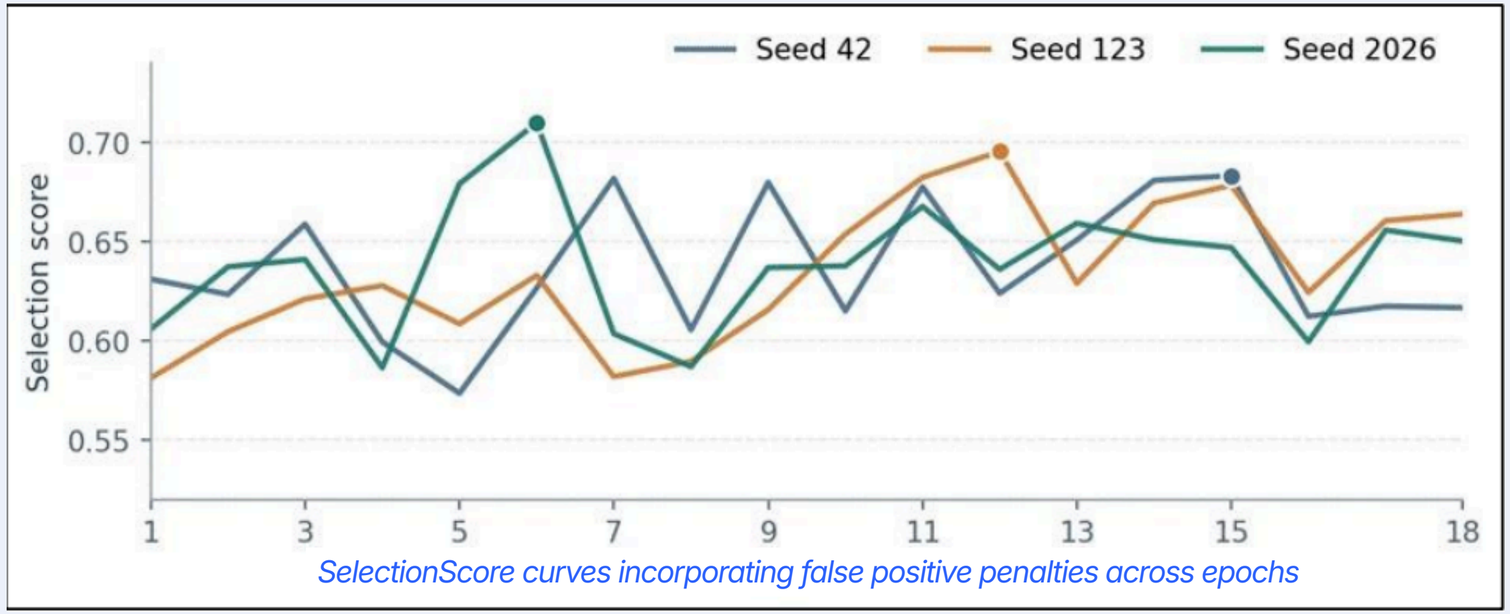
Balance overlap quality against false-positive masks on healthy skin

Seed	Epoch	Val. Dice	NegativeFPR	Score
42	15	69.24%	6.12%	0.6832
123	12	70.00%	3.06%	0.6954
2026	6	71.29%	2.04%	0.7099

SelectionScore
=
Validation Dice − 0.15 × NegativeFPR

Selected: seed 2026, epoch 6

- Validation IoU: 60.66%
- FPR measured on 98 held-out None images



SelectionScore curves incorporating false positive penalties across epochs

Segmentation model

Curriculum sampling strategy

Keep the source mix fixed while expanding the eligible positive image pool

Four strategies evaluated

Uniform · Easy-to-Hard · Hard-to-Easy · Random Curriculum

- The pool reaches full size by approximately 75% of training.
- Pseudo-mask difficulty ranking showed no consistent benefit across seeds.

25% manual / 55% pseudo-mask / 20% None

Identical source sampling ratios across experiments

Curriculum: 30% → 100% of eligible positive data

Curriculum vs. Manual-only baseline

Same preprocessing and evaluation metrics for both checkpoints

Higher test Dice on 9 of 13 images

Evaluation	Metric	Manual-only v6	Curriculum v5
Validation (n=21)	Mean Dice	64.78%	71.30%
Validation (n=21)	Mean IoU	53.07%	60.67%
Test (n=13)	Mean Dice	65.92%	72.15%
Test (n=13)	Mean IoU	54.64%	60.14%

Segmentation model

Interpretation and limitations

Internal test performance does not establish real-world reliability

What the results show

- Binary suspected-burn mask
- Test Mean Dice: 72.15%
- Test Mean IoU: 60.14%
- Manual test set: 13 images

What remains Limited

- Small manual validation and test sets
- Uneven annotation boundary precision
- No burn-degree prediction per pixel
- No quantitative external validation
- None labels are not clinical pixel masks

Next steps: expand specialist-verified masks and external evaluation

Experimental Model Selection

Criterion	Classification	Segmentation
Candidates	5 CNN/Transformer architectures	4 decoder/encoder architectures
Evaluation Set for Selection	462 validation images, 4 classes	21 manual validation masks + 98 holdout None images
Input, Batch Size	224 × 224, 32	320 × 320, 8
Training Budget	15 epochs per seed	18 epochs × 80 steps/epoch per seed
Seeds	42, 123, 2026	42, 123, 2026
Selection Metric	Validation Macro-F1	Validation Dice - 0.15 × NegativeFPR
Initialization	Pretrained (timm)	Pretrained encoder, random curriculum

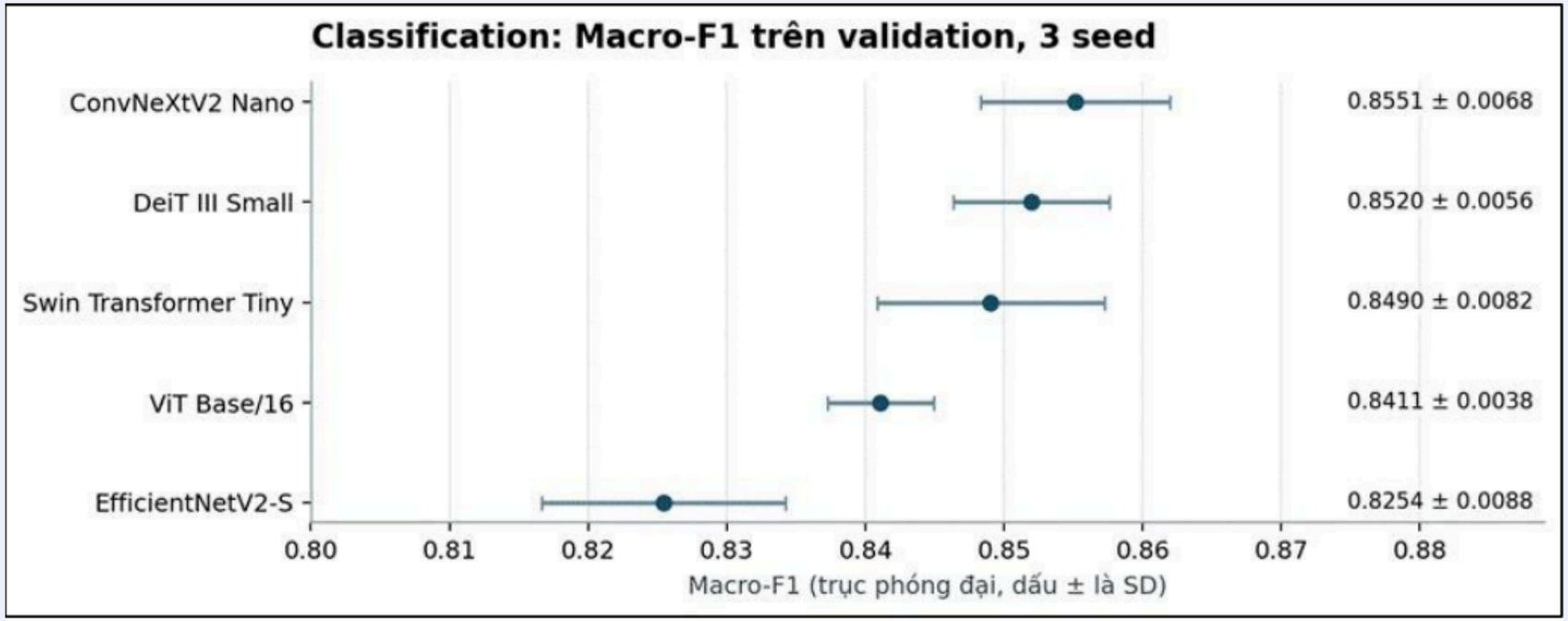
- **Objective:** Select architectures through controlled experiments and validation performance.
- **27 training runs:** 5 classification models + 4 segmentation models × 3 seeds (42, 123, 2026).
- **Hardware:** NVIDIA A100-SXM4-40GB GPU.
- Within each task, models shared the same data splits, training budget, and checkpoint criteria.
- Results were reported as mean ± standard deviation across seeds.

Selection criteria: Validation Macro-F1 for classification; Validation Dice – 0.15 × NegativeFPR for segmentation.

Classification Architecture Selection

Decision

Select ConvNeXtV2 Nano for its leading average performance and compact size.



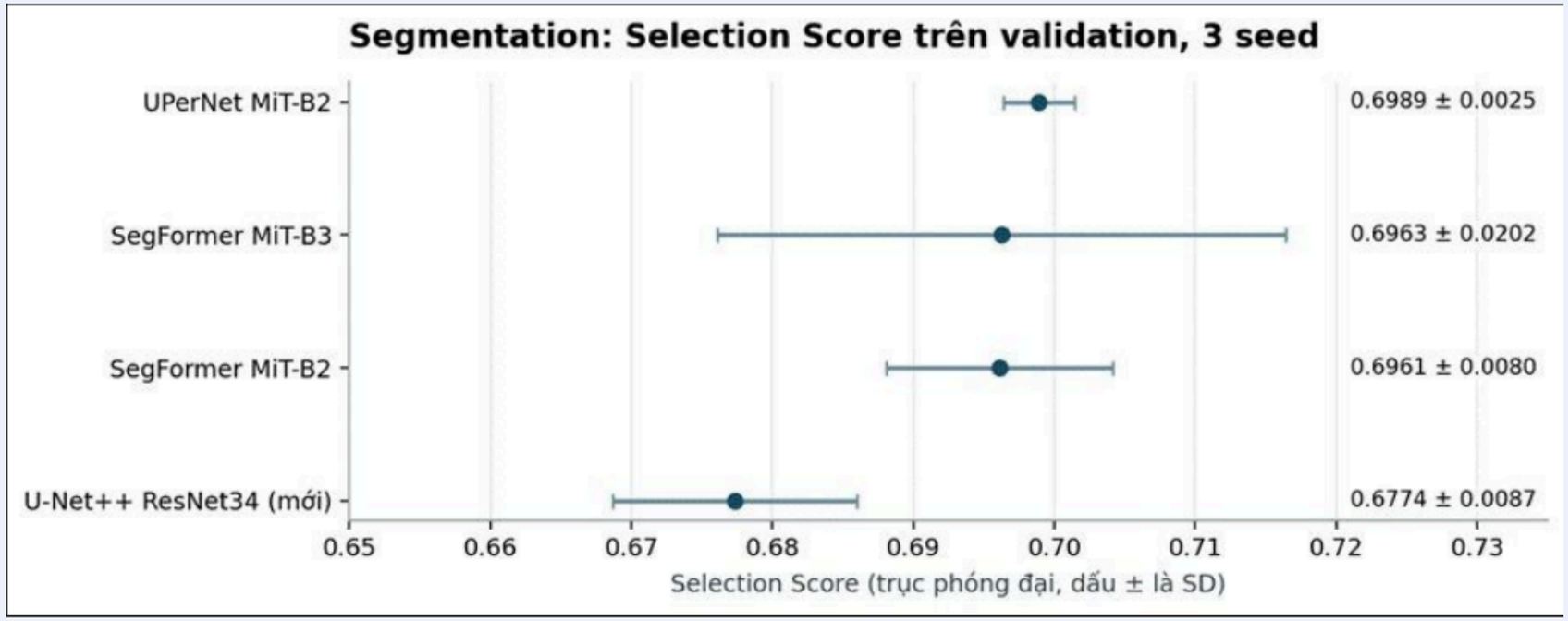
Average Macro-F1 and standard deviation across three seeds; the horizontal axis is magnified to show small differences

- Compared **ConvNeXtV2 Nano, DeiT III Small, Swin Tiny, ViT Base/16, and EfficientNetV2-S.**
- **ConvNeXtV2 Nano ranked first:** validation Macro-F1 85.51% ± 0.68 percentage points.
- DeiT III Small (85.20%) and Swin Tiny (84.90%) achieved similar average scores.
- ConvNeXtV2 also had the fewest parameters: 14.99M.

Classification Architecture Selection

Finding

UPerNet offered the strongest validation balance of overlap quality, false-positive control, and consistency across seeds.



Average validation Selection Score and standard deviation across three seeds; horizontal axis is magnified to distinguish architectures

- Compared **U-Net++ ResNet-34, SegFormer MiT-B2, SegFormer MiT-B3, and UPerNet MiT-B2** using Random Curriculum.
- **UPerNet MiT-B2 ranked first:** validation SelectionScore 0.6989 ± 0.0025.
- It achieved 0% NegativeFPR on 98 held-out None images at all three selected checkpoints.
- SegFormer MiT-B2 had slightly higher average Dice (70.02%), but false positives reduced its SelectionScore.

Final model decision

- **Classification:** Selected **ConvNeXtV2 Nano** for the highest average validation Macro-F1 across five architectures and compact parameter count.
- **Segmentation:** Retained **Curriculum U-Net++ ResNet-34** (random_curriculum, seed 2026) for slightly higher test Mean Dice, zero empty masks across 13 test images, and a verified demo pipeline.
- **Alternative:** Preserved UPerNet MiT-B2 as a strong candidate due to minimal Mean Dice difference and superior performance on several other metrics.

ConvNeXtV2 predicts image-level labels; U-Net++ generates binary masks for visualization and contradiction checking.

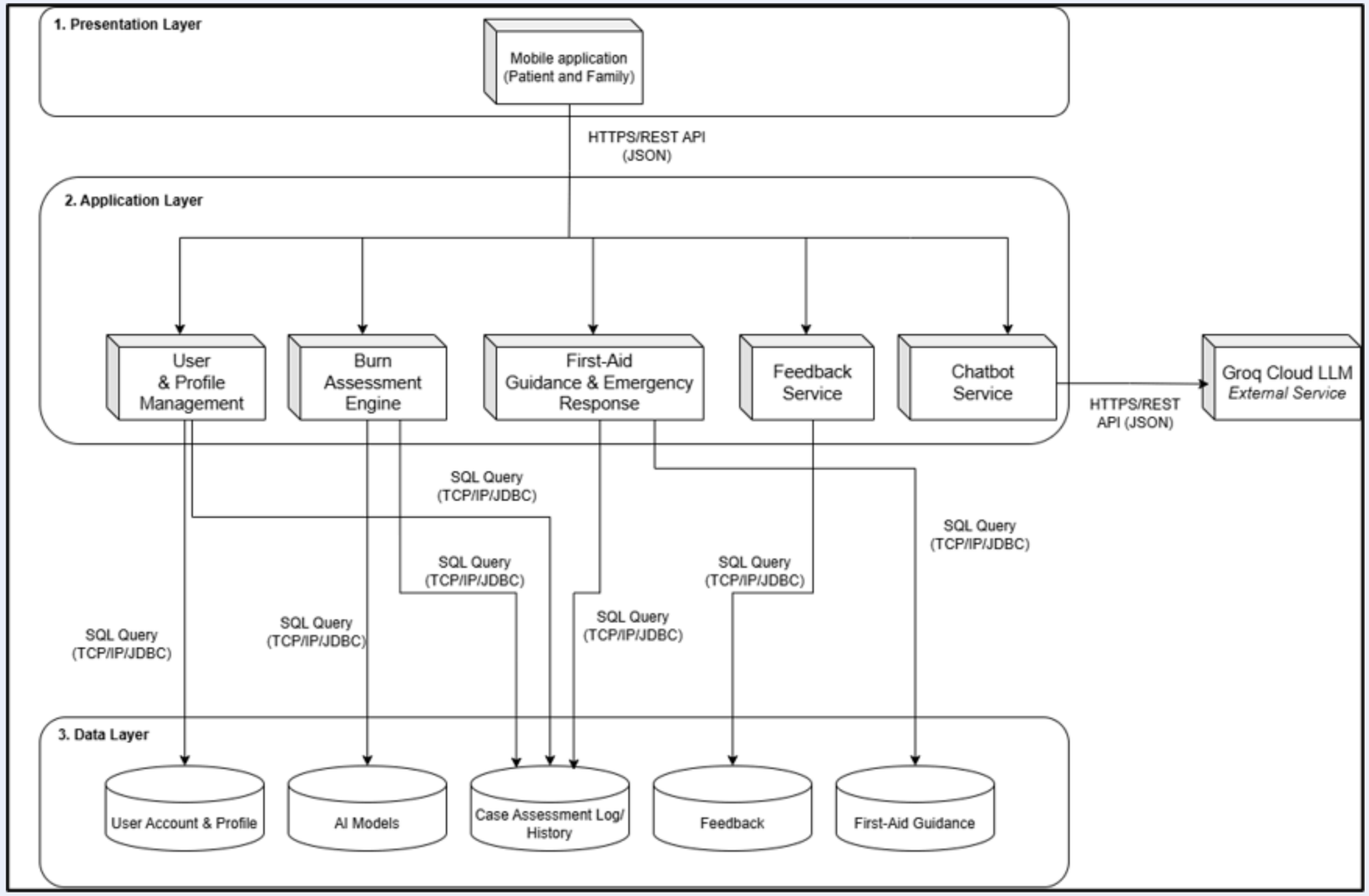
Large language model & RAG

Large Language Model (LLM) combined with Retrieval-Augmented Generation (RAG) serves as an intelligent medical assistant.

- **LLM:** Open-source LLaMA 3.3 70B Versatile, deployed via Groq API, with millisecond response times.
- **Knowledge base:** 18 reputable textbooks and professional guidelines, including Military Medical Academy and Ministry of Health documents.
- **Data processing:** 2,274 text chunks stored in knowledge_base_burns_full.json.
- **Retrieval:** BM25 and vector embeddings retrieve 3–5 relevant chunks to construct LLM prompt context.

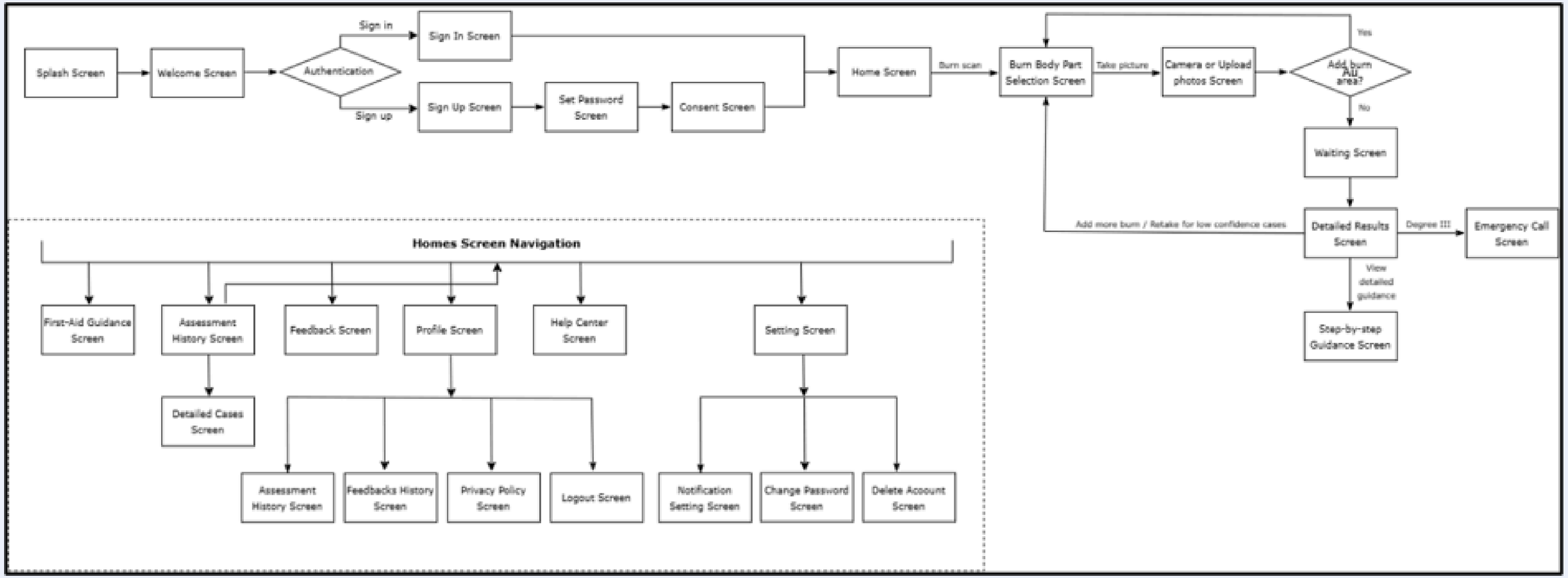
Medical accuracy: Removed unverified professional titles, such as “board-certified,” from system prompts.

3-tier architecture diagram



Burn Aid

Navigation flowchart



UI design



Burn Aid

UI design

Demo

System Limitations



Clinical scope

- Thermal burns caused by fire only;
- Not yet an age-adjusted for children under 18;



Features

- One photo per assessment;
- Local SQLite; no doctor review screen or location sharing.



AI and imaging

- Performance of models is affected by the quality of images taken in real situations
- Tiny scattered burns missed; chatbot needs internet.

Short-Term Plan

1. Multi-injury assessment
2. Lund–Browder chart for under-18s
3. Doctor review of low-confidence cases
4. Chemical and electrical first aid

Medium & Long-Term Plan

1. Local clinical dataset, Vietnamese skin
2. Build a local clinical dataset for better reflecting
3. GIS: nearest facility, pre-hospital info

Thank You


09:45
Choose Burn Body Part
Front View
Back View
TAP ON A NUMBERED REGION
SELECTED REGION
Forearm, Front, Right
Prof. Le The Trung Standard:
1.5% TBSA base for this view
ESTIMATED TBSA FOR REGION
0.75% TBSA
Burn affected extent in region:
50%
0%
20%
40%
60%
80%
100%
Reset Selection
Take Picture Now